

# Number System and its conversion

## SY. Btech\_EXTC



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# Content

**Digital Number  
System**

**Types based on  
radix: Binary,  
Decimal, Octal,  
Hexadecimal,  
BCD, XS-3,  
Gray**

**Number-base  
conversion**

**Complements**

**Signed numbers**

**Parity: Error  
detection and  
correction**

# Learning Outcomes

Why number systems?

Types of number system

Able to convert from one form to another

# Digital Number System

Symbolic number system uses Roman numerals (I = 1, X = 10, L = 50, C = 100 and M = 1000)

Weighted Position Number System

Numbers: communicate, perform tasks, quantify, measure etc..

## Number system types

Binary

Octal

Decimal

Hexadecimal

## Codes

Binary code

Excess-3 code

Gray code

Error detection and correction codes

# Number representation

- Any number system is collection of digits
- It has two parts: Integer and Fraction
- Separated by a radix point
- $(N)_b = d_{n-1} d_{n-2} \dots d_2 d_1 d_0 . d_{-1} d_{-2} \dots d_{-m}$
- Where N= a number
- b = radix or base of number system
- m= number of digits in fractional portion
- n = number of digits in integer portion
- $d_{n-1}$  = most significant digit (MSB)
- $d_{-m}$  = least significant digit (LSB)

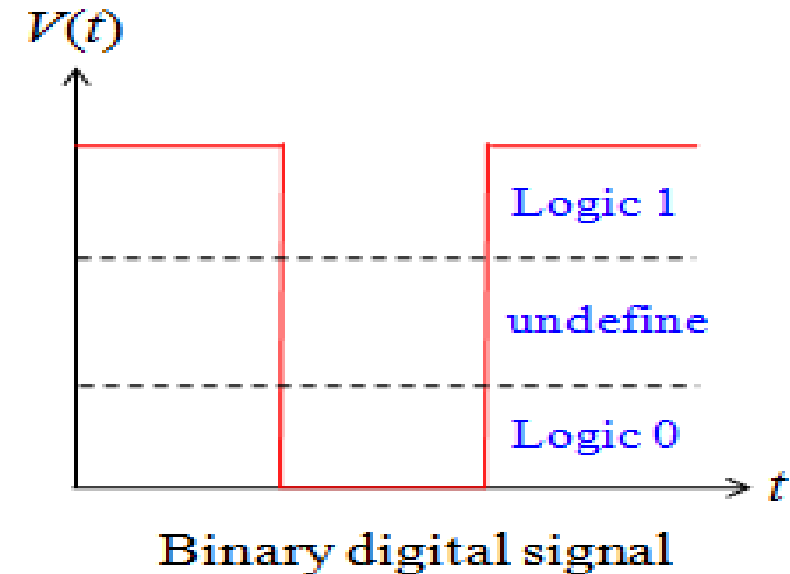
# Number- Base System

| Name        | Radix | Digits                          |
|-------------|-------|---------------------------------|
| Binary      | 2     | 0,1                             |
| Octal       | 8     | 0,1,2,3,4,5,6,7                 |
| Decimal     | 10    | 0,1,2,3,4,5,6,7,8,9             |
| Hexadecimal | 16    | 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F |

- The six letters (in addition to the 10 integers) in hexadecimal represent: 10, 11, 12, 13, 14, and 15, respectively.

# Digital signal- Binary

- An information represented by electrical quantity.
- For digital systems, the variable is discrete values.
  - Two level binary values.
- Noise tolerance is high.
- Easy data transmission.
- Easy to build binary circuits.

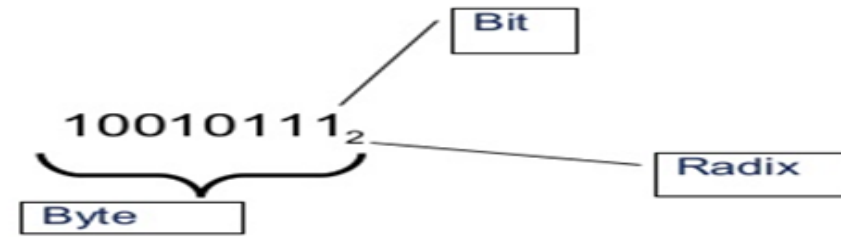


# Digital signal- Binary

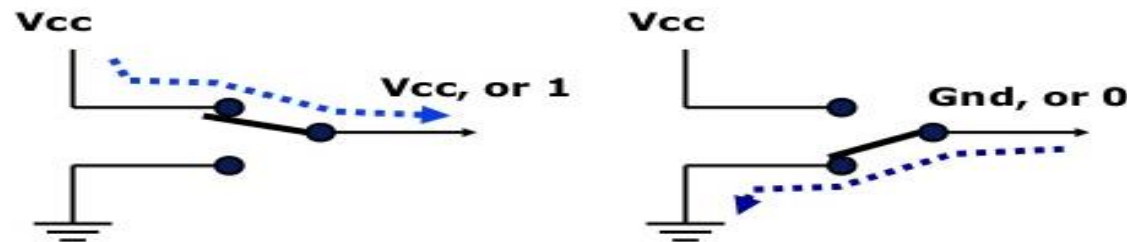
Bit: Single binary digit

Group of Binary digits

- Nibble (4)
- Byte (8)
- Word (16)
- Quad (32)

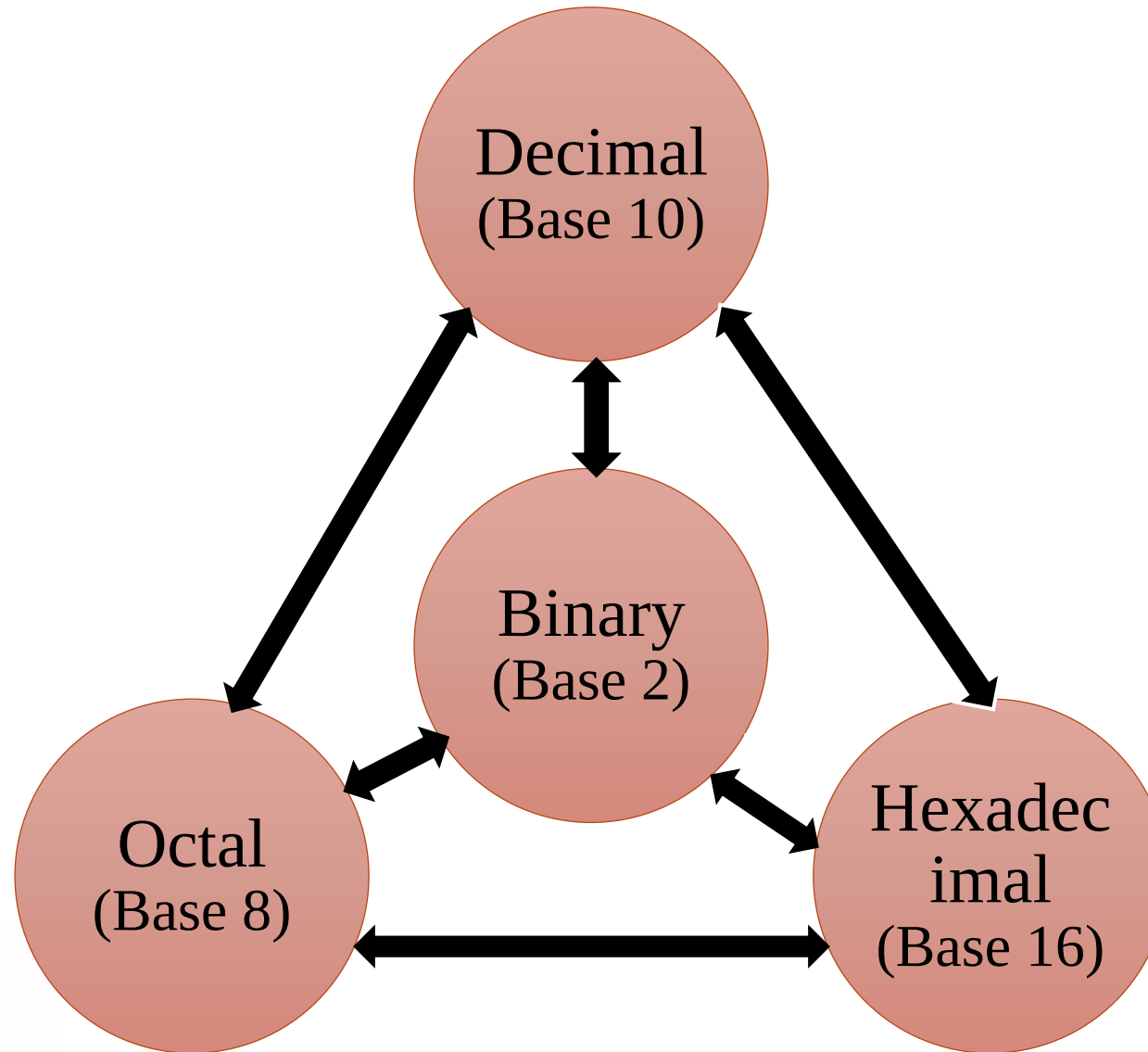


Here is a simple switch used to provide a logic value





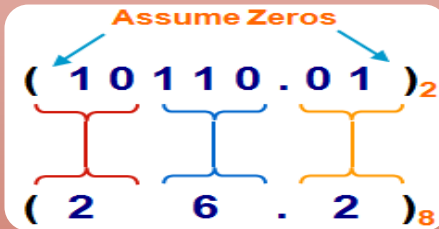
# Conversion between number bases



# Binary Number System

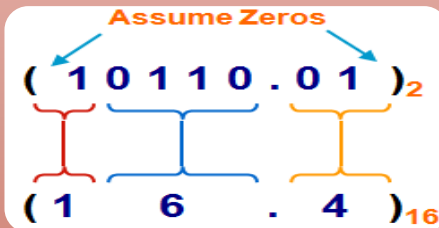
## Binary to Decimal conversion

$$1*2^2+0*2^1+1*2^0+0*2^{-1}+1*2^{-2}=(5.25)_{10}$$



## Binary to Octal conversion

- Group the binary digits into groups of three
- $(11011001)_2 = (011)(011)(001) = (331)_8$



## Binary to Hexadecimal conversion

- Group the binary digits into groups of four
- $(11011001)_2 = (1101)(1001) = (D9)_{16}$

# Decimal Number System

## Decimal to Binary conversion:

- Divide the number by the 'Base' (=2)
  - Obtain the remainder (either 0 or 1) as a coefficient
  - Note the quotient and repeat the division
- 
- Multiply the number by the 'Base' (=2)
  - Obtain the integer (either 0 or 1) as a coefficient
  - Take the resultant fraction and repeat the multiplication

Example:  $(13)_{10}$

|          | Quotient                                     | Remainder | Coefficient |
|----------|----------------------------------------------|-----------|-------------|
| $13/2 =$ | 6                                            | 1         | $a_0 = 1$   |
| $6/2 =$  | 3                                            | 0         | $a_1 = 0$   |
| $3/2 =$  | 1                                            | 1         | $a_2 = 1$   |
| $1/2 =$  | 0                                            | 1         | $a_3 = 1$   |
| Answer:  | $(13)_{10} = (a_3 a_2 a_1 a_0)_2 = (1101)_2$ |           |             |
|          |                                              | MSB       | LSB         |

Example:  $(0.625)_{10}$

|               | Integer                                                 | Fraction | Coefficient  |
|---------------|---------------------------------------------------------|----------|--------------|
| $0.625 * 2 =$ | 1                                                       | . 25     | $a_{-1} = 1$ |
| $0.25 * 2 =$  | 0                                                       | . 5      | $a_{-2} = 0$ |
| $0.5 * 2 =$   | 1                                                       | . 0      | $a_{-3} = 1$ |
| Answer:       | $(0.625)_{10} = (0.a_{-1} a_{-2} a_{-3})_2 = (0.101)_2$ |          |              |
|               |                                                         | MSB      | LSB          |

# Decimal Number System

## Decimal to Octal conversion:

- Divide the number by the 'Base' (=8)
- Obtain the remainder as a coefficient
- Note the quotient and repeat the division

Example:  $(175)_{10}$

|             | Quotient | Remainder | Coefficient |
|-------------|----------|-----------|-------------|
| $175 / 8 =$ | 21       | 7         | $a_0 = 7$   |
| $21 / 8 =$  | 2        | 5         | $a_1 = 5$   |
| $2 / 8 =$   | 0        | 2         | $a_2 = 2$   |

Answer:  $(175)_{10} = (a_2 a_1 a_0)_8 = (257)_8$

- Multiply the number by the 'Base' (=8)
- Obtain the integer as a coefficient
- Take the resultant fraction and repeat the multiplication

Example:  $(0.3125)_{10}$

|                | Integer | Fraction | Coefficient  |
|----------------|---------|----------|--------------|
| $0.3125 * 8 =$ | 2       | . 5      | $a_{-1} = 2$ |
| $0.5 * 8 =$    | 4       | . 0      | $a_{-2} = 4$ |

Answer:  $(0.3125)_{10} = (0.a_{-1} a_{-2} a_{-3})_8 = (0.24)_8$

# Decimal Number System

## Decimal to Hexadecimal conversion:

- Divide the number by the 'Base' (=16)
- Obtain the remainder as a coefficient
- Note the quotient and repeat the division

|                           | Quotient | Remainder |
|---------------------------|----------|-----------|
| $678 \div 16$             | 42       | 6         |
| $42 \div 16$              | 2        | A         |
| $2 \div 16$               | 0        | 2         |
| $(678)_{10} = (2A6)_{16}$ |          |           |

- Multiply the number by the 'Base' (=16)
- Obtain the integer as a coefficient
- Take the resultant fraction and repeat the multiplication

$$\begin{aligned}0.342_{10} &= ?_{16} \\0.342 \times 16 &= 5.472 \quad (.5_{16}) \\0.472 \times 16 &= 7.552 \quad (.57_{16}) \\0.552 \times 16 &= 8.832 \quad (.578_{16}) \\0.832 \times 16 &= 13.312 \quad (.578D_{16}) \\0.342_{10} &\approx 0.578D_{16}\end{aligned}$$

# Signed Binary Number System

- Three methods to represent signed binary number
  - Sign-magnitude method
  - One's complement method
  - Two's complement method

## 1. Sign-Magnitude Method:

- Additional bit is used as sign bit
- Sign bit is placed as the Most Significant Bit(MSB)
- Bit 0 for +ve number
- Bit 1 for – ve number
- E.g. 8 bit signed number 01000100 represents +ve number
- Its magnitude is  $(1000100)_2 = +(68)_{10}$

# Sign magnitude representation

- ‘+’ sign before a number indicates it as a positive number.
- ‘-’ sign before a number indicates it as a negative number.
- Less compatible with computers.
- Replace ‘+’ sign by 0 and ‘-’ by 1:-
  - $(+1100101)_2 = (01100101)_2$
  - $(+101.001)_2 = (0101.001)_2$
  - $(-10010)_2 = (110010)_2$
  - $(-110.101)_2 = (1110.101)_2$

# Complements

- 1's Complement (*Diminished Radix Complement*)

All '0's become '1's

All '1's become '0's

Example:  $(10110000)_2 = (01001111)_2$

$$\begin{array}{r} 10110000 \\ + 01001111 \\ \hline 11111111 \end{array}$$

- 2's Complement (*Radix Complement*)

➤ Take 1's complement then add 1

**OR**

➤ Toggle all bits to the left of the first '1' from the right

$$\begin{array}{r} 10110000 \\ 01001111 \\ + \phantom{00000000} 1 \\ \hline 01010000 \end{array}$$



# 1's complement representation method for signed binary numbers

- Replacement of bit 0 by bit 1 and bit 1 by 0 in the given number is 1's complement of that number
- E.g. 1's complement of following numbers will be found as follows:

If given number is 0100111001

Its 1's complement will be 1011000110

- For n bit number, the max positive number which can be represented in 1's complement form is  $(2^{n-1}-1)$  and max -ve number is  $-(2^{n-1}-1)$
- Represent the following decimal numbers in 1's complement form
  - (+7) , (-7), (+8), (-8), (+15),(-15)

# 2's complement representation for signed binary numbers with conversion rules

- If LSB of the number is 1, its 2's complement is obtained by changing each 0 to 1 and each 1 to 0 except LSB
- If LSB of the number is 0, 2's complement is obtained by scanning the number from LSB to MSB bit by bit and retaining the bits as they are up to including occurrence of first 1 and complement all other bits.
- Examples – Find 2's complement of following numbers using conversion rules

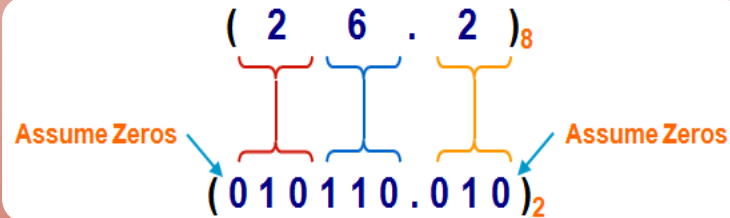
1) 01001110  $\rightarrow$  2's complement is 10110010

2) 00110101  $\rightarrow$  2's complement is 11001011

# Range of numbers in data representation

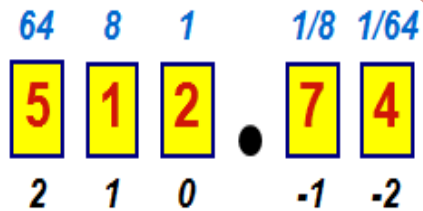
| Sign-Magnitude method            | 1's complement data representation | 2's complement data representation |
|----------------------------------|------------------------------------|------------------------------------|
| $-(2^{n-1}-1)$ to $+(2^{n-1}-1)$ | $-(2^{n-1}-1)$ to $+(2^{n-1}-1)$   | $-(2^n)$ to $+(2^{n-1}-1)$         |

# Octal Number System



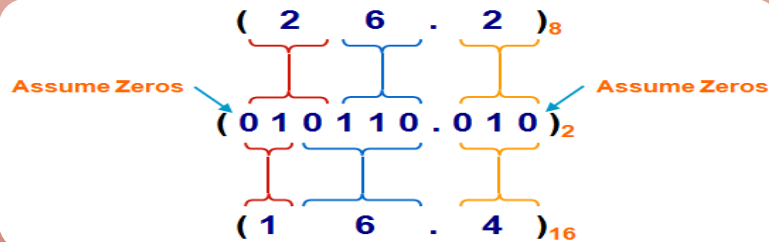
## Octal to binary conversion

Group the binary digits into groups of three  
 $(010110010)_2 = (010)(110)(010) = (26.2)_8$



## Octal to Decimal conversion

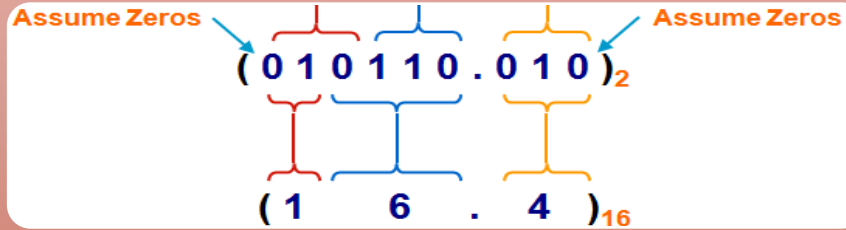
$$5 \times 8^2 + 1 \times 8^1 + 2 \times 8^0 + 7 \times 8^{-1} + 4 \times 8^{-2} = (330.9375)_{10}$$



## Octal to Hexadecimal conversion

Convert to Binary as an intermediate step  
 $(010110010)_2 = (0001)(0110)(0100) = (16.4)_{16}$

# Hexadecimal Number System



## Hexadecimal to binary conversion

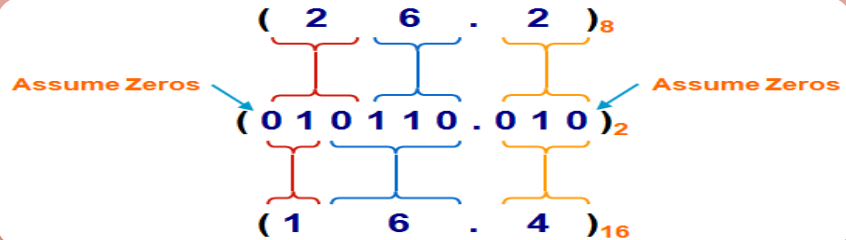
Group the binary digits into groups of four

$$(16.4)_{16} = (000101100100)_2 = (0001)\ (0110)\ (0100)$$

|     |    |   |      |       |
|-----|----|---|------|-------|
| 256 | 16 | 1 | 1/16 | 1/256 |
| 1   | E  | 5 | 7    | A     |
| 2   | 1  | 0 | -1   | -2    |

## Hexadecimal to Decimal conversion

$$\bullet\ 1*16^2 + 14*16^1 + 5*16^0 + 7*16^{-1} + 10*16^{-2} = (485.4765625)_{10}$$



## Hexadecimal to Octal conversion

Convert to Binary as an intermediate step

$$(010110010)_2 = (010)\ (110)\ (010) = (26.2)_8$$

# Decimal, Binary, Octal and Hexadecimal Number System

| Decimal | Binary | Octal | Hexadecimal |
|---------|--------|-------|-------------|
| 00      | 0000   | 00    | 0           |
| 01      | 0001   | 01    | 1           |
| 02      | 0010   | 02    | 2           |
| 03      | 0011   | 03    | 3           |
| 04      | 0100   | 04    | 4           |
| 05      | 0101   | 05    | 5           |
| 06      | 0110   | 06    | 6           |
| 07      | 0111   | 07    | 7           |
| 08      | 1000   | 10    | 8           |
| 09      | 1001   | 11    | 9           |
| 10      | 1010   | 12    | A           |
| 11      | 1011   | 13    | B           |
| 12      | 1100   | 14    | C           |
| 13      | 1101   | 15    | D           |
| 14      | 1110   | 16    | E           |
| 15      | 1111   | 17    | F           |

# Binary coded decimal (BCD) Code

- A number with k decimal digits will require 4k bits in BCD.
- Example: Decimal 396 is represented in BCD with 12bits as 0011 1001 0110, with each group of 4 bits representing one decimal digit.
- A decimal number in BCD is the same as its equivalent binary number only when the number is between 0 and 9.
- The binary combinations 1010 through 1111 are not used.

## Binary-Coded Decimal (BCD)

| Decimal Symbol | BCD Digit |
|----------------|-----------|
| 0              | 0000      |
| 1              | 0001      |
| 2              | 0010      |
| 3              | 0011      |
| 4              | 0100      |
| 5              | 0101      |
| 6              | 0110      |
| 7              | 0111      |
| 8              | 1000      |
| 9              | 1001      |

## BCD codes continued...

- Also known as 8-4-2-1 code. 8, 4, 2, 1 are the weights of the four bits of the binary code of each decimal digit similar to straight binary numbers
- **So, this is a weighted code.**



## Excess-3 code

- **Excess-3 code : Another type of BCD code**
- Add '0011' (3) to all the 8421 code or natural BCD code of each digit.
- E.g. Decimal 2 is coded as  $0010 + 0011 = 0101$
- Excess-3 is not a weighted code

# Types of BCD Codes

## Four Different Binary Codes for the Decimal Digits

| Decimal Digit           | BCD 8421 | 2421 | Excess-3 | 8, 4, -2, -1 |
|-------------------------|----------|------|----------|--------------|
| 0                       | 0000     | 0000 | 0011     | 0000         |
| 1                       | 0001     | 0001 | 0100     | 0111         |
| 2                       | 0010     | 0010 | 0101     | 0110         |
| 3                       | 0011     | 0011 | 0110     | 0101         |
| 4                       | 0100     | 0100 | 0111     | 0100         |
| 5                       | 0101     | 1011 | 1000     | 1011         |
| 6                       | 0110     | 1100 | 1001     | 1010         |
| 7                       | 0111     | 1101 | 1010     | 1001         |
| 8                       | 1000     | 1110 | 1011     | 1000         |
| 9                       | 1001     | 1111 | 1100     | 1111         |
| Unused bit combinations | 1010     | 0101 | 0000     | 0001         |
|                         | 1011     | 0110 | 0001     | 0010         |
|                         | 1100     | 0111 | 0010     | 0011         |
|                         | 1101     | 1000 | 1101     | 1100         |
|                         | 1110     | 1001 | 1110     | 1101         |
|                         | 1111     | 1010 | 1111     | 1110         |



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# Gray Code

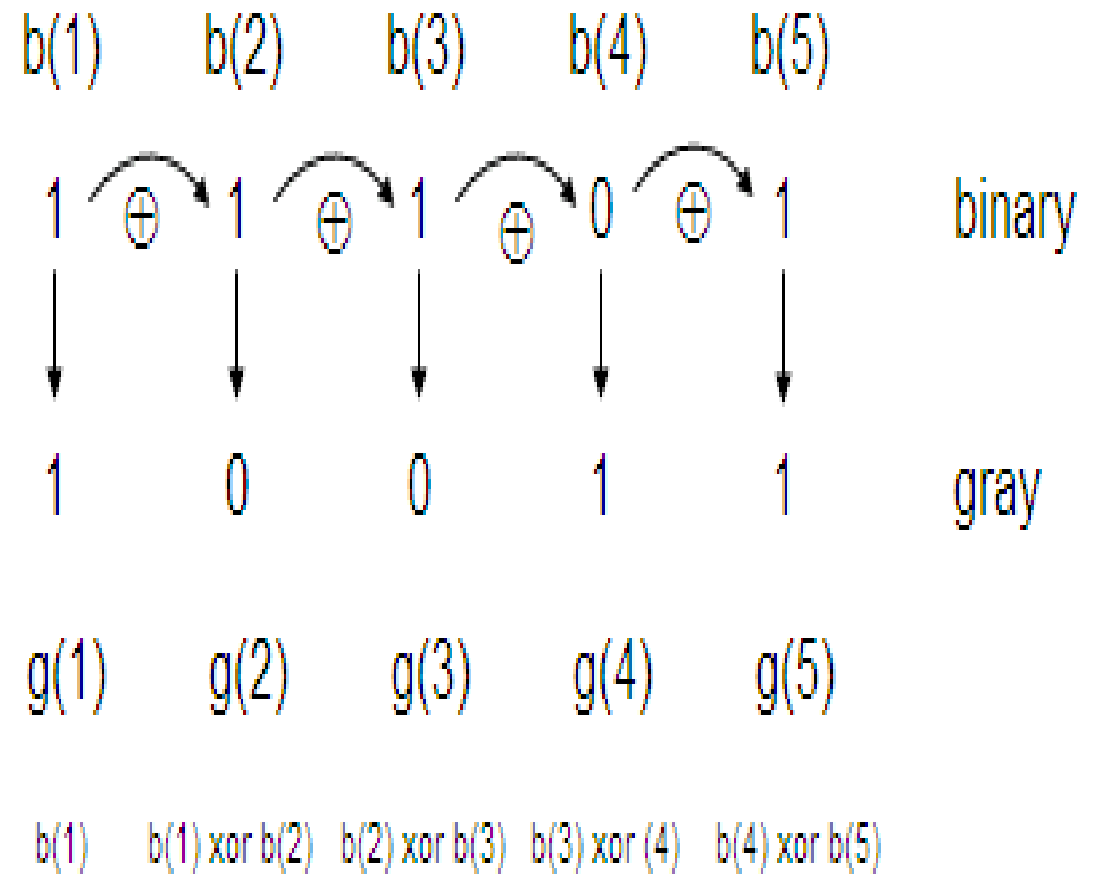
- Only bit in the code group changes while going from one number to the next.
- Error detection.
- Representation of analog data.
- Low power design.

*Gray Code*

| Gray Code | Decimal Equivalent |
|-----------|--------------------|
| 0000      | 0                  |
| 0001      | 1                  |
| 0011      | 2                  |
| 0010      | 3                  |
| 0110      | 4                  |
| 0111      | 5                  |
| 0101      | 6                  |
| 0100      | 7                  |
| 1100      | 8                  |
| 1101      | 9                  |
| 1111      | 10                 |
| 1110      | 11                 |
| 1010      | 12                 |
| 1011      | 13                 |
| 1001      | 14                 |
| 1000      | 15                 |

# Gray code continued...

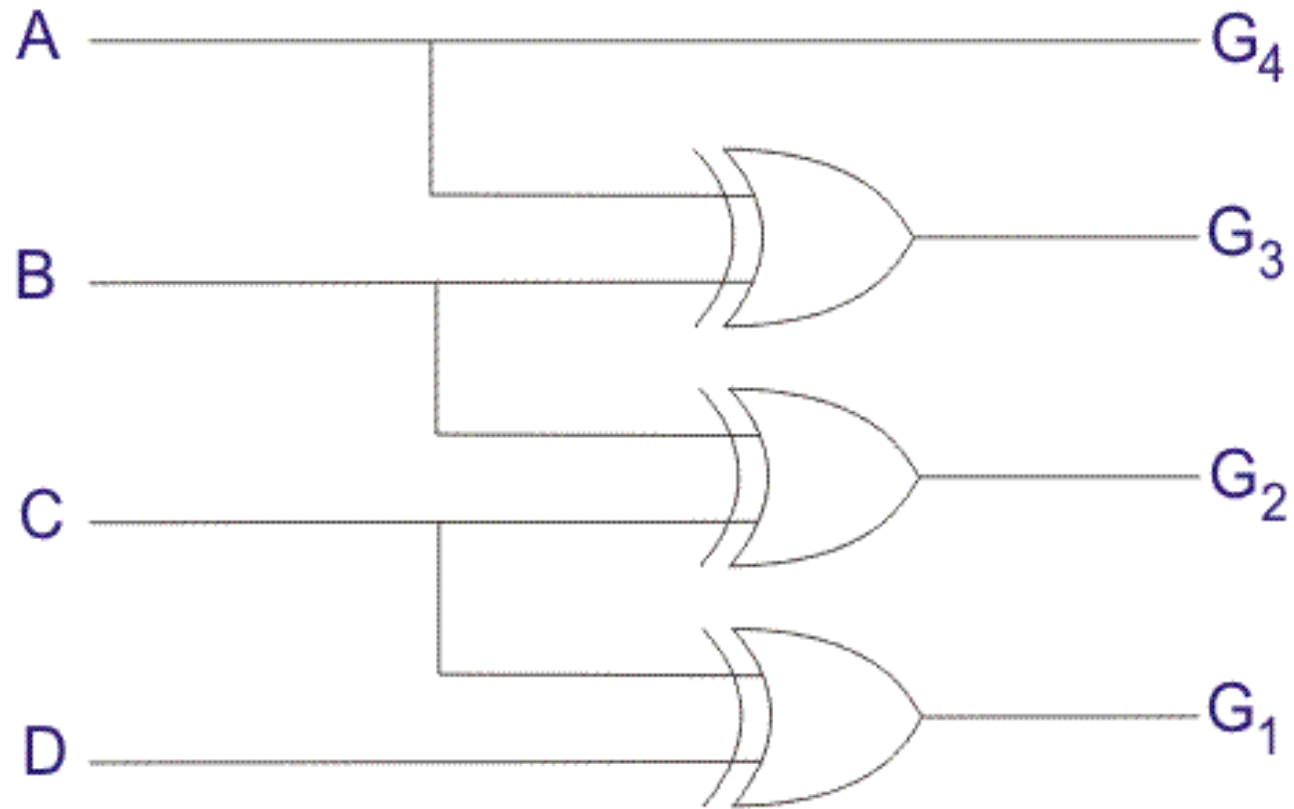
- Gray code is not weighted code
- It is also a reflected code.
- Procedure for Binary to Gray code conversion



## Binary to Gray code table

| Decimal | Binary Code | Gray Code |
|---------|-------------|-----------|
| 0       | 0000        | 0000      |
| 1       | 0001        | 0001      |
| 2       | 0010        | 0011      |
| 3       | 0011        | 0010      |
| 4       | 0100        | 0110      |
| 5       | 0101        | 0111      |
| 6       | 0110        | 0101      |
| 7       | 0111        | 0100      |
| 8       | 1000        | 1100      |
| 9       | 1001        | 1101      |
| 10      | 1010        | 1111      |
| 11      | 1011        | 1110      |
| 12      | 1100        | 1010      |
| 13      | 1101        | 1011      |
| 14      | 1110        | 1001      |
| 15      | 1111        | 1000      |

# Binary to Gray converter circuit



Logic Circuit for Binary to Gray Code Converter

# Need for coding

- Information sent over a noisy channel is likely to be distorted.
- Information is coded to facilitate : Efficient transmission , Error detection and correction.
- Coding and Decoding process.
- A parity bit is an extra bit included with a message to make the total number of 1's either even or odd.

A = 1000001  
T = 1010100

**With even parity    With odd parity**

01000001  
11010100

11000001  
01010100

# Octal code

- 3 bit binary code
- Each of octal digits 0 through 7 is coded into 3 bit straight binary number
- E.g. for octal digit 4 ,code is 100
- This code is useful for binary inputs in digital computers, microprocessors, etc.



# Hexadecimal Code

- It is 4 bit binary code
- E.g. for hexadecimal digit 8 code is 1000. For C  $(12)_{10}$  code is 1100
- This code is useful for binary inputs in digital computers, microprocessors, etc.

# Excercise

- Find the decimal equivalents of the following binary numbers:

(a)  $(101)_2$  (b)  $(1001)_2$  (c)  $(10.011)_2$   
[(a)  $(5)_{10}$  (b)  $(9)_{10}$  (c)  $(3.0375)_{10}$ ]

- What are the decimal equivalents of the following binary numbers ?

(a)  $(1111)_2$  (b)  $(10100)_2$  (c)  $(11011001)_2$  (d)  $(10011001)_2$   
[(a)  $(15)_{10}$  (b)  $(20)_{10}$  (c)  $(109)_{10}$  (d)  $(153)_{10}$ ]

- Express the following binary numbers into their equivalent decimal numbers :

(a)  $(11.01)_2$  (b)  $(101.11)_2$  (c)  $(110.01)_2$   
[(a)  $(3.25)_{10}$  (b)  $(5.75)_{10}$  (c)  $(6.25)_{10}$ ]

- Convert the following decimal numbers into their binary equivalents:

(a)  $(25)_{10}$  (b)  $(125)_{10}$  (c)  $(0.85)_{10}$   
[ (a)  $(11001)_2$  (b)  $(1111101)_2$  (c)  $(0.110110)_2$ ]

- What are the binary equivalents of the following decimal numbers ?

(a)  $(27)_{10}$  (b)  $(92)_{10}$  (c)  $(64)_{10}$   
[(a)  $(11011)_2$  (b)  $(1011100)_2$  (c)  $(1000000)_2$ ]

- Convert the following real numbers to the binary numbers:

(i)  $(12.0)_{10}$  (ii)  $(25.0)_{10}$  (iii)  $(0.125)_{10}$   
[(i)  $(1100)_2$  (ii)  $(11001)_2$  (iii)  $(0.001)_2$ ]



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*Thank You*

